

Rahul Rawat

rahulrawat2r@gmail.com • 015563603340 • linkedin.com/in/rahulrawat2r •
github.com/rahulrawat20022002 • Mannheim, Germany

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Robert Bosch GmbH

Hiring Team

Application: Master Thesis, Graph Based Question Answering and Retrieval Augmented Generation Systems

Dear Hiring Team,

I am applying for the Master Thesis on Graph Based Question Answering and Retrieval Augmented Generation Systems at Robert Bosch GmbH. The proposal sits at the intersection of two areas I have been actively working in, modern GraphRAG systems and rigorous scientific evaluation, and it maps cleanly onto my Master's programme at SRH Heidelberg.

For the past year I have been building a modular Retrieval Augmented Generation orchestrator around Llama 3.1 8b served through Groq. I designed a router that classifies user intent into local vector retrieval from a ChromaDB store, external DuckDuckGo search, or direct conversational logic, and layered a stateful memory agent on top so context survives across turns. That work gives me a very concrete starting point for exploring how symbolic reasoning over RDF and OWL knowledge graphs, combined with SPARQL retrieval, can improve faithfulness and answer accuracy compared to purely embedding based baselines.

The research posture matches how I already work. In my CreditIQ project I compared classical fair machine learning approaches, diagnosed a hidden intersectional bias with SHAP, designed a four way age by gender threshold matrix, and documented the fairness accuracy trade off as a regulator defensible decision under EU AI Act Annex III and GDPR Article 22. In my Bachelor Thesis I compared six classifiers on a clinical dataset with 10 fold cross validation and per model confusion matrices, wrote the results as an IEEE style paper, and included an honest limitations section. I bring the same evaluation discipline to any new comparison against state of the art baselines on Bosch internal and public benchmarks.

I would be delighted to discuss the thesis proposal in more detail and to share my code repositories. Thank you for considering my application.

Kind regards,

Rahul Rawat